

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with @BENCH adt.fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT	SoA	Fold passes			Map passes		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
OctTree	16	9	0.3737	0.3270	1.14 ×	0.3030	0.3514	0.86×
Compiler	9	8	0.1330	0.0395	3.36 ×	0.1604	0.1782	0.90×
DBQuery	15	13	0.1261	0.0861	1.47 ×	0.1374	0.1657	0.83×
DecisionTree	7	6	3.865	2.927	1.32 ×	—	—	—
DomTree	15	14	0.1452	0.0613	2.37 ×	0.0581	0.0784	0.74×
KDTree	17	16	0.1879	0.1386	1.36 ×	—	—	—
LinearListReduction	11	11	0.0296	0.0050	5.91 ×	—	—	—
List	2	2	0.1305	0.1118	1.17 ×	0.2562	0.3395	0.75×
MonoTree	3	2	0.0177	0.0225	0.79×	0.0454	0.0427	1.06×
ObjectGraph	5	4	0.0976	0.0806	1.21 ×	0.1444	0.1602	0.90×
PiecewiseFunctions	8	7	0.1428	0.0945	1.51 ×	0.2387	0.2592	0.92×
TernaryTree	5	3	0.0277	0.0257	1.08×	0.0830	0.0779	1.07×
Trie	10	9	0.0663	0.0326	2.03 ×	0.1497	0.1695	0.88×
ColorOctree	25	18	0.0919	0.0763	1.21 ×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
OctTree	1.03×	0.75×	1.47 ×	0.42×	0.24×	0.48×	3.54 ×
Compiler	1.68 ×	0.52×	3.80 ×	0.66×	14.63 ×	0.31×	6.53 ×
DBQuery	1.11 ×	0.47×	1.34 ×	0.37×	8.63 ×	0.22×	6.19 ×
DecisionTree	1.32 ×	0.59×	18.32 ×	2.25 ×	1.01×	2.31 ×	36.98 ×
DomTree	1.61 ×	0.57×	3.02 ×	1.06×	20.67 ×	0.17×	10.16 ×
KDTree	1.36 ×	0.77×	1.22 ×	0.83×	0.22×	0.84×	4.92 ×
LinearListReduction	5.93 ×	0.90×	32.53 ×	5.30 ×	40.67 ×	5.30 ×	11.54 ×
List	0.81×	0.80×	3.41 ×	0.70×	0.11×	0.61×	1.46 ×
MonoTree	0.97×	0.85×	0.52×	0.73×	0.26×	0.76×	1.22 ×
ObjectGraph	1.00×	0.75×	6.77 ×	0.88×	0.55×	0.52×	2.85 ×
PiecewiseFunctions	1.12 ×	0.60×	2.42 ×	1.23 ×	1.93 ×	0.42×	4.39 ×
TernaryTree	1.14 ×	0.79×	0.34×	0.41×	0.22×	0.85×	1.35 ×
Trie	1.13 ×	0.56×	3.82 ×	0.67×	3.52 ×	0.14×	6.34 ×
Geomean	1.32 ×	0.67×	2.79 ×	0.87×	1.46 ×	0.57×	4.73 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
OctTree	2.6e+09/ 2.5e+09	8.2e+09 /1.1e+10	1.5e+07/ 1.0e+07	3.7e+05 /8.8e+05	6.1e+04 /2.5e+05	1.2e+04 /2.4e+04	6.4e+07/ 1.8e+07
Compiler	9.8e+08/ 5.8e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.4e+06	2.2e+05 /3.4e+05	1.3e+06/ 8.7e+04	1.1e+04 /3.5e+04	5.0e+07/ 7.7e+06
DBQuery	9.7e+08/ 8.7e+08	1.1e+09 /2.2e+09	4.0e+06/ 3.0e+06	1.6e+05 /4.3e+05	8.4e+05/ 9.8e+04	6.8e+03 /3.1e+04	2.0e+07/ 3.3e+06
DecisionTree	2.0e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.4e+08/ 2.9e+07	1.0e+05/ 4.6e+04	2.1e+06/ 2.1e+06	1.0e+05/ 4.4e+04	1.1e+09/ 2.9e+07
DomTree	8.6e+08/ 5.3e+08	1.2e+09 /2.2e+09	1.3e+07/ 4.2e+06	1.2e+05/ 1.1e+05	1.1e+06/ 5.1e+04	5.2e+03 /3.1e+04	5.6e+07/ 5.5e+06
KDTree	9.5e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.5e+07/ 1.2e+07	8.8e+02 /1.1e+03	2.2e+04 /9.9e+04	8.8e+02 /1.0e+03	4.4e+07/ 8.9e+06
LinearListReduction	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	2.3e+02/ 4.3e+01	2.8e+05/ 7.0e+03	2.3e+02/ 4.3e+01	1.2e+07/ 1.1e+06
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.2e+06/ 2.7e+06	4.2e+05 /6.0e+05	2.2e+04 /2.0e+05	6.2e+03 /1.0e+04	4.9e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.4e+05 /4.5e+05	3.2e+04 /4.4e+04	3.4e+03 /1.3e+04	1.7e+03 /2.2e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.7e+08 /8.7e+08	3.0e+09 /4.0e+09	9.2e+06/ 1.4e+06	2.3e+05 /2.6e+05	2.9e+04 /5.3e+04	9.4e+03 /1.8e+04	2.3e+07/ 8.2e+06
PiecewiseFunctions	1.3e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.4e+07/ 5.8e+06	4.9e+05/ 4.0e+05	1.8e+05/ 9.2e+04	1.9e+04 /4.7e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.5e+05 /1.6e+06	6.0e+04 /1.5e+05	4.4e+03 /2.0e+04	2.3e+03 /2.7e+03	3.3e+06/ 2.4e+06
Trie	6.6e+08/ 5.9e+08	1.2e+09 /2.2e+09	9.2e+06/ 2.4e+06	1.7e+05 /2.6e+05	1.7e+05/ 4.9e+04	6.0e+03 /4.2e+04	2.6e+07/ 4.0e+06

Table 4: Baseline Gibbon comparison (times only). Shown variants: AoS-mut, AoS-imm, SoA-mut. Times are median per iteration (s). **Bold** marks the fastest time across shown variants.

Program	Am	Ai	Sm
Compiler	0.2934	1.095	0.2178
DBQuery	0.2636	0.3319	0.2517
DecisionTree	3.865	5.701	2.927
DomTree	0.2033	0.3446	0.1397
KDTree	0.1879	0.2435	0.1386
LinearListReduction	0.0296	0.1330	0.0050
List	0.3867	<i>OOM</i>	0.4514
MonoTree	0.0631	0.1418	0.0652
ObjectGraph	0.2420	0.4854	0.2408
Oct:sumMass	0.0339	0.0444	0.0274
Oct:sumEnergy	0.0377	0.0540	0.0319
Oct:countActive	0.0359	0.0460	0.0290
Oct:countParticles	0.0348	0.0454	0.0276
Oct:barnesHutPotential	0.0429	0.0585	0.0393
Oct:fmPotential	0.0965	0.0967	0.0956
Oct:scaleEnergy	0.1501	0.1522	0.1755
Oct:clearFlags	0.1529	0.1510	0.1759
PiecewiseFunctions	0.3815	0.5633	0.3538
TernaryTree	0.1107	0.1446	0.1035
Trie	0.2160	0.2931	0.2021
ColorOctree	0.0919	0.1185	0.0763

Table 5: Baseline Gibbon comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm
Compiler	1.35 $\times$	3.73 $\times$	5.03 $\times$
DBQuery	1.05 $\times$	1.26 $\times$	1.32 $\times$
DecisionTree	1.32 $\times$	1.47 $\times$	1.95 $\times$
DomTree	1.46 $\times$	1.70 $\times$	2.47 $\times$
KDTree	1.36 $\times$	1.30 $\times$	1.76 $\times$
LinearListReduction	5.91 $\times$	4.50 $\times$	26.59 $\times$
List	0.86 $\times$	–	–
MonoTree	0.97 $\times$	2.25 $\times$	2.18 $\times$
ObjectGraph	1.01 $\times$	2.01 $\times$	2.02 $\times$
Oct:sumMass	1.24 $\times$	1.31 $\times$	1.62 $\times$
Oct:sumEnergy	1.18 $\times$	1.43 $\times$	1.69 $\times$
Oct:countActive	1.24 $\times$	1.28 $\times$	1.59 $\times$
Oct:countParticles	1.26 $\times$	1.31 $\times$	1.64 $\times$
Oct:barnesHutPotential	1.09 $\times$	1.36 $\times$	1.49 $\times$
Oct:fmnPotential	1.01 $\times$	1.00 $\times$	1.01 $\times$
Oct:scaleEnergy	0.86 $\times$	1.01 $\times$	0.87 $\times$
Oct:clearFlags	0.87 $\times$	0.99 $\times$	0.86 $\times$
PiecewiseFunctions	1.08 $\times$	1.48 $\times$	1.59 $\times$
TernaryTree	1.07 $\times$	1.31 $\times$	1.40 $\times$
Trie	1.07 $\times$	1.36 $\times$	1.45 $\times$
ColorOctree	1.21 $\times$	1.29 $\times$	1.55 $\times$

Table 6: Per-pass performance for `OctTree`, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
barnesHutPotential	F	11/16	31%	0.0429 $\pm$ 0.00002	0.0585 $\pm$ 0.00016	0.0393$\pm$0.00010	1.09 $\times$	1.36 $\times$	1.49 $\times$
clearFlags	M	15/16	6%	0.1529 $\pm$ 0.00022	0.1510$\pm$0.00209	0.1759 $\pm$ 0.00166	0.87 $\times$	0.99 $\times$	0.86 $\times$
countActive	F	10/16	38%	0.0359 $\pm$ 0.00001	0.0460 $\pm$ 0.00005	0.0290$\pm$0.00010	1.24 $\times$	1.28 $\times$	1.59 $\times$
countParticles	F	8/16	50%	0.0348 $\pm$ 0.00001	0.0454 $\pm$ 0.00002	0.0276$\pm$0.00003	1.26 $\times$	1.31 $\times$	1.64 $\times$
fmnPotential	F	12/16	25%	0.0965 $\pm$ 0.00003	0.0967 $\pm$ 0.00005	0.0956$\pm$0.00130	1.01 $\times$	1.00 $\times$	1.01 $\times$
scaleEnergy	M	16/16	0%	0.1501$\pm$0.00009	0.1522 $\pm$ 0.00212	0.1755 $\pm$ 0.00087	0.86 $\times$	1.01 $\times$	0.87 $\times$
sumEnergy	F	12/16	25%	0.0377 $\pm$ 0.00095	0.0540 $\pm$ 0.00002	0.0319$\pm$0.00002	1.18 $\times$	1.43 $\times$	1.69 $\times$
sumMass	F	10/16	38%	0.0339 $\pm$ 0.00002	0.0444 $\pm$ 0.00094	0.0274$\pm$0.00022	1.24 $\times$	1.31 $\times$	1.62 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0442 $\pm$ 0.00004	0.0537 $\pm$ 0.00005	0.0334$\pm$0.00007	1.32 $\times$	1.22 $\times$	1.61 $\times$
quantizationErrorProxy	F	10/25	60%	0.0477 $\pm$ 0.00001	0.0648 $\pm$ 0.00152	0.0428$\pm$0.00001	1.11 $\times$	1.36 $\times$	1.51 $\times$
Total				0.6767	0.7666	0.6784	1.00 $\times$	1.13 $\times$	1.13 $\times$
Geomean							1.11 $\times$	1.22 $\times$	1.35 $\times$

Table 7: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.2e+08/ 2.0e+08	8.2e+08 /1.0e+09	7.1e+04 /5.9e+05	1.1e+02/ 6.5e+01	2.0e+03 /2.3e+04	1.1e+02/ 6.7e+01	6.2e+06/ 2.3e+06
clearFlags	M	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	8.6e+05 /2.6e+06	1.6e+05 /4.5e+05	9.2e+03 /3.8e+04	5.6e+03 /1.1e+04	5.4e+06/ 2.5e+06
countActive	F	1.8e+08/ 1.5e+08	5.8e+08/ 5.8e+08	8.2e+05/ 3.8e+05	1.7e+02/ 3.9e+01	3.9e+03 /1.5e+04	1.7e+02/ 4.3e+01	6.2e+06/ 3.2e+05
countParticles	F	1.8e+08/ 1.4e+08	5.7e+08/ 5.7e+08	9.6e+05/ 2.1e+05	1.6e+02/ 6.5e+01	3.5e+03 /7.6e+03	1.6e+02/ 6.9e+01	6.1e+06/ 6.2e+04
fmnPotential	F	4.9e+08/ 4.9e+08	1.6e+09 /1.8e+09	2.7e+04 /7.2e+05	4.0e+02/ 3.0e+02	1.8e+03 /2.5e+04	4.0e+02/ 3.0e+02	4.4e+06/ 2.2e+06
scaleEnergy	M	3.5e+08 /4.4e+08	9.1e+08 /1.9e+09	1.3e+06 /2.7e+06	2.1e+05 /4.3e+05	8.6e+03 /3.7e+04	4.2e+03 /1.2e+04	5.4e+06/ 2.5e+06
sumEnergy	F	1.9e+08/ 1.6e+08	7.7e+08 /8.9e+08	3.1e+05 /5.9e+05	4.3e+02/ 6.8e+01	2.1e+03 /2.3e+04	4.3e+02/ 6.8e+01	6.1e+06/ 2.7e+06
sumMass	F	1.7e+08/ 1.4e+08	5.7e+08 /6.3e+08	8.0e+05/ 2.1e+05	1.3e+02/ 4.4e+01	2.7e+03 /9.8e+03	1.3e+02/ 4.8e+01	6.3e+06/ 1.2e+06
paletteEntriesQuantized	F	2.2e+08/ 1.7e+08	6.7e+08 /7.6e+08	5.4e+06/ 1.3e+06	3.7e+02/ 1.4e+02	1.2e+04 /4.0e+04	3.7e+02/ 1.5e+02	9.2e+06/ 1.4e+06
quantizationErrorProxy	F	2.4e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.4e+06/ 9.2e+05	7.0e+01 /2.9e+02	1.6e+04 /3.3e+04	7.0e+01 /3.0e+02	9.2e+06/ 2.9e+06
Total		2.6e+09/2.5e+09	8.2e+09/1.1e+10	1.5e+07/1.0e+07	3.7e+05/8.8e+05	6.1e+04/2.5e+05	1.2e+04/2.4e+04	6.4e+07/1.8e+07

Table 8: Per-pass performance for `Compiler`, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
blockCountPass	F	2/9	78%	0.0127 $\pm$ 0.00000	0.0790 $\pm$ 0.00097	0.0019$\pm$0.00000	6.65 $\times$	6.20 $\times$	41.23 $\times$
branchStatsPass	F	2/9	78%	0.0123 $\pm$ 0.00001	0.0795 $\pm$ 0.00077	0.0036$\pm$0.00000	3.46 $\times$	6.46 $\times$	22.33 $\times$
castInstCountPass	F	3/9	67%	0.0194 $\pm$ 0.00055	0.0795 $\pm$ 0.00002	0.0013$\pm$0.00000	15.03 $\times$	4.10 $\times$	61.64 $\times$
has cycle	F	4/9	56%	0.0246 $\pm$ 0.00011	0.0937 $\pm$ 0.00003	0.0172$\pm$0.00017	1.43 $\times$	3.81 $\times$	5.44 $\times$
instCountPass	F	2/9	78%	0.0128 $\pm$ 0.00001	0.0802 $\pm$ 0.00005	0.0023$\pm$0.00000	5.61 $\times$	6.27 $\times$	35.21 $\times$
latencyModelPass	F	3/9	67%	0.0127 $\pm$ 0.00000	0.0802 $\pm$ 0.00003	0.0030$\pm$0.00000	4.26 $\times$	6.31 $\times$	26.86 $\times$
memoryOpStatsPass	F	3/9	67%	0.0125 $\pm$ 0.00001	0.0794 $\pm$ 0.00116	0.0038$\pm$0.00001	3.26 $\times$	6.36 $\times$	20.75 $\times$
stripSideEffectsPass	M	7/9	22%	0.0800$\pm$0.00014	0.1875 $\pm$ 0.00028	0.0906 $\pm$ 0.00112	0.88 $\times$	2.34 $\times$	2.07 $\times$
targetReturnPass	M	9/9	0%	0.0804$\pm$0.00017	0.1861 $\pm$ 0.00102	0.0877 $\pm$ 0.00008	0.92 $\times$	2.31 $\times$	2.12 $\times$
throughputModelPass	F	3/9	67%	0.0128 $\pm$ 0.00000	0.0796 $\pm$ 0.00080	0.0030$\pm$0.00000	4.25 $\times$	6.22 $\times$	26.48 $\times$
verifyIR	F	9/9	0%	0.0131 $\pm$ 0.00003	0.0706 $\pm$ 0.00007	0.0034$\pm$0.00005	3.80 $\times$	5.39 $\times$	20.46 $\times$
Total				0.2934	1.095	0.2178	1.35 $\times$	3.73 $\times$	5.03 $\times$
Geomean						3.30 $\times$	4.76 $\times$	15.71 $\times$	

Table 9: Per-pass PAPI counters for `Compiler` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.5e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.3e+06/ 6.4e+03	1.6e+02/ 2.4e+01	2.3e+05/ 1.1e+03	1.6e+02/ 2.5e+01	4.6e+06/ 9.8e+02
branchStatsPass	F	6.2e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.5e+05/ 2.1e+04	3.5e+02/ 4.5e+01	5.2e+04/ 3.2e+03	3.5e+02/ 4.6e+01	4.6e+06/ 4.2e+05
castInstCountPass	F	9.8e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.3e+04 /2.2e+04	5.9e+01/ 1.9e+01	8.3e+02 /1.2e+03	6.0e+01/ 1.9e+01	4.6e+06/ 3.3e+03
has cycle	F	1.2e+08/ 8.7e+07	1.1e+08 /1.5e+08	2.4e+06/ 3.0e+05	2.0e+02/ 1.2e+02	1.5e+05/ 2.2e+04	2.0e+02/ 1.2e+02	5.1e+06/ 2.4e+06
instCountPass	F	6.5e+07/ 1.1e+07	5.6e+07 /5.7e+07	1.3e+06/ 1.4e+04	1.3e+02/ 3.3e+01	2.3e+05/ 1.1e+03	1.4e+02/ 3.3e+01	4.6e+06/ 1.3e+03
latencyModelPass	F	6.4e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.2e+04	1.4e+02/ 3.8e+01	2.1e+05/ 3.7e+03	1.4e+02/ 3.9e+01	4.6e+06/ 5.3e+05
memoryOpStatsPass	F	6.3e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.8e+05/ 2.7e+04	1.1e+02/ 4.6e+01	3.4e+04/ 3.1e+03	1.2e+02/ 4.8e+01	4.6e+06/ 4.3e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.4e+06 /1.4e+06	1.2e+05 /1.6e+05	7.5e+03 /2.0e+04	5.9e+03 /1.8e+04	4.1e+06/ 1.3e+06
targetReturnPass	M	1.5e+08 /1.8e+08	1.7e+08 /6.3e+08	1.6e+06/ 1.5e+06	1.0e+05 /1.8e+05	6.3e+03 /2.4e+04	3.2e+03 /1.6e+04	4.0e+06/ 1.7e+06
throughputModelPass	F	6.5e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	1.8e+02/ 4.5e+01	2.1e+05/ 3.5e+03	1.8e+02/ 4.4e+01	4.7e+06/ 5.4e+05
verifyIR	F	6.6e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.4e+05/ 1.7e+04	2.2e+02/ 1.2e+02	1.4e+05/ 3.9e+03	2.1e+02/ 1.2e+02	4.6e+06/ 4.5e+05
Total		9.8e+08/5.8e+08	9.9e+08/1.9e+09	1.3e+07/3.4e+06	2.2e+05/3.4e+05	1.3e+06/8.7e+04	1.1e+04/3.5e+04	5.0e+07/7.7e+06

Table 10: Per-pass performance for DBQuery, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
clearQueryFlags	M	14/15	7%	0.0690$\pm$0.00009	0.0760 $\pm$ 0.00156	0.0820 $\pm$ 0.00121	0.84 $\times$	1.10$\times$	0.93 $\times$
countJoins	F	3/15	80%	0.0197 $\pm$ 0.00002	0.0290 $\pm$ 0.00012	0.0110$\pm$0.00001	1.80$\times$	1.47$\times$	2.65$\times$
filterSelectivitySkew	F	5/15	67%	0.0197 $\pm$ 0.00077	0.0291 $\pm$ 0.00001	0.0124$\pm$0.00001	1.59$\times$	1.48$\times$	2.35$\times$
hashJoinPressure	F	5/15	67%	0.0204 $\pm$ 0.00001	0.0332 $\pm$ 0.00001	0.0124$\pm$0.00000	1.65$\times$	1.63$\times$	2.68$\times$
scaleCosts	M	15/15	0%	0.0684$\pm$0.00023	0.0761 $\pm$ 0.00009	0.0836 $\pm$ 0.00025	0.82 $\times$	1.11$\times$	0.91 $\times$
sumCost	F	6/15	60%	0.0196 $\pm$ 0.00001	0.0289 $\pm$ 0.00157	0.0141$\pm$0.00007	1.39$\times$	1.48$\times$	2.06$\times$
sumMemory	F	6/15	60%	0.0198 $\pm$ 0.00001	0.0291 $\pm$ 0.00001	0.0144$\pm$0.00002	1.37$\times$	1.47$\times$	2.02$\times$
sumRows	F	6/15	60%	0.0269 $\pm$ 0.00001	0.0305 $\pm$ 0.00002	0.0218$\pm$0.00002	1.23$\times$	1.13$\times$	1.39$\times$
Total				0.2636	0.3319	0.2517	1.05 $\times$	1.26$\times$	1.32$\times$
Geomean						1.29$\times$	1.34$\times$	1.73$\times$	

Table 11: Per-pass PAPI counters for DBQuery (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08/2.1e+08	1.9e+08/7.2e+08	2.6e+05/1.2e+06	8.0e+04/2.5e+05	3.8e+04/ 2.4e+04	2.5e+03/1.3e+04	2.4e+06/ 9.3e+05
countJoins	F	1.0e+08/ 5.5e+07	7.4e+07/ 7.1e+07	7.1e+05/ 2.6e+04	4.7e+02/ 6.1e+01	1.7e+05/ 2.1e+03	4.8e+02/ 5.3e+01	2.7e+06/ 1.7e+03
filterSelectivitySkew	F	9.9e+07/ 6.3e+07	1.0e+08/1.1e+08	5.2e+05/ 7.5e+04	1.8e+02/ 5.4e+01	1.2e+05/ 6.1e+03	1.8e+02/ 5.7e+01	2.6e+06/ 2.3e+05
hashJoinPressure	F	1.0e+08/ 6.3e+07	8.4e+07/9.9e+07	7.0e+05/ 9.2e+04	1.9e+02/ 8.3e+01	1.8e+05/ 7.2e+03	2.0e+02/ 8.7e+01	2.7e+06/ 1.1e+05
scaleCosts	M	1.6e+08/2.2e+08	2.2e+08/7.5e+08	2.6e+05/1.3e+06	7.8e+04/1.8e+05	3.5e+04/ 2.7e+04	2.9e+03/1.7e+04	2.5e+06/ 1.1e+06
sumCost	F	9.9e+07/ 7.1e+07	9.0e+07/1.2e+08	6.7e+05/ 8.0e+04	2.0e+02/ 5.9e+01	1.5e+05/ 1.1e+04	2.0e+02/ 6.0e+01	2.7e+06/ 3.3e+05
sumMemory	F	1.0e+08/ 7.3e+07	9.0e+07/1.2e+08	5.7e+05/ 8.6e+04	2.8e+02/ 8.0e+01	1.3e+05/ 1.2e+04	2.8e+02/ 8.5e+01	2.6e+06/ 3.3e+05
sumRows	F	1.4e+08/ 1.1e+08	2.1e+08/2.3e+08	2.5e+05/ 9.6e+04	1.6e+02/ 8.0e+01	2.2e+04/ 7.7e+03	1.6e+02/ 8.3e+01	2.3e+06/ 2.9e+05
Total		9.7e+08/8.7e+08	1.1e+09/2.2e+09	4.0e+06/3.0e+06	1.6e+05/4.3e+05	8.4e+05/9.8e+04	6.8e+03/3.1e+04	2.0e+07/3.3e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
classify Batch	F	5/7	29%	3.320 $\pm$ 0.0065	4.330 $\pm$ 0.0060	2.584$\pm$0.0039	1.29$\times$	1.30$\times$	1.68$\times$
classify Depth	F	4/7	43%	0.0473 $\pm$ 0.00001	0.0617 $\pm$ 0.00001	0.0365$\pm$0.00103	1.29$\times$	1.30$\times$	1.69$\times$
classify tree	F	5/7	29%	0.0474 $\pm$ 0.00001	0.0617 $\pm$ 0.00002	0.0365$\pm$0.00007	1.30$\times$	1.30$\times$	1.69$\times$
countFeatureUses	F	3/7	57%	0.0505 $\pm$ 0.00001	0.1414 $\pm$ 0.00019	0.0322$\pm$0.00002	1.57$\times$	2.80$\times$	4.39$\times$
countLeaves	F	2/7	71%	0.0477 $\pm$ 0.00003	0.1401 $\pm$ 0.00002	0.0250$\pm$0.00001	1.91$\times$	2.93$\times$	5.60$\times$
countNodes	F	2/7	71%	0.0477 $\pm$ 0.00003	0.1403 $\pm$ 0.00295	0.0261$\pm$0.00002	1.82$\times$	2.94$\times$	5.37$\times$
countSmallLeaves	F	3/7	57%	0.0514 $\pm$ 0.00082	0.1361 $\pm$ 0.00204	0.0322$\pm$0.00003	1.59$\times$	2.65$\times$	4.22$\times$
inferenceCost	F	2/7	71%	0.0508 $\pm$ 0.00001	0.1391 $\pm$ 0.00121	0.0288$\pm$0.00001	1.76$\times$	2.74$\times$	4.82$\times$
sumImpurity	F	3/7	57%	0.0505 $\pm$ 0.00001	0.1390 $\pm$ 0.00141	0.0319$\pm$0.00010	1.58$\times$	2.75$\times$	4.35$\times$
sumPathLengths	F	3/7	57%	0.0521 $\pm$ 0.00001	0.1385 $\pm$ 0.00021	0.0340$\pm$0.00001	1.53$\times$	2.66$\times$	4.08$\times$
sumSamples	F	3/7	57%	0.0485 $\pm$ 0.00001	0.1342 $\pm$ 0.00116	0.0323$\pm$0.00002	1.50$\times$	2.77$\times$	4.15$\times$
treeDepth	F	2/7	71%	0.0514 $\pm$ 0.00001	0.1388 $\pm$ 0.00198	0.0280$\pm$0.00107	1.84$\times$	2.70$\times$	4.96$\times$
Total				3.865	5.701	2.927	1.32$\times$	1.47$\times$	1.95$\times$
Geomean						1.57$\times$	2.29$\times$	3.60$\times$	

Table 13: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.7e+10/ 1.3e+10	3.3e+10/5.9e+10	4.6e+08/ 2.7e+07	9.4e+04/ 4.0e+04	1.8e+06/1.9e+06	9.1e+04/ 3.7e+04	9.2e+08/ 1.7e+07
classify Depth	F	2.4e+08/ 1.9e+08	4.7e+08/8.5e+08	6.5e+06/ 3.7e+05	1.5e+03/ 6.4e+02	2.0e+04/2.8e+04	1.5e+03/ 6.1e+02	1.3e+07/ 2.8e+05
classify tree	F	2.4e+08/ 1.9e+08	4.7e+08/8.5e+08	6.5e+06/ 3.7e+05	9.6e+02/ 6.0e+02	2.5e+04/2.8e+04	9.6e+02/ 5.7e+02	1.3e+07/ 2.8e+05
countFeatureUses	F	2.6e+08/ 1.6e+08	7.5e+08/7.8e+08	7.3e+06/ 2.5e+05	1.5e+03/ 9.6e+02	3.3e+04/ 1.3e+04	1.5e+03/ 9.6e+02	1.5e+07/ 2.0e+06
countLeaves	F	2.4e+08/ 1.3e+08	6.0e+08/6.1e+08	7.3e+06/ 1.1e+05	8.1e+02/ 3.9e+02	5.3e+04/ 8.0e+03	8.1e+02/ 3.9e+02	1.4e+07/ 3.3e+05
countNodes	F	2.4e+08/ 1.3e+08	6.1e+08/6.2e+08	7.3e+06/ 1.0e+05	6.1e+02/ 2.4e+02	4.9e+04/ 8.2e+03	6.1e+02/ 2.4e+02	1.4e+07/ 3.4e+05
countSmallLeaves	F	2.6e+08/ 1.6e+08	7.1e+08/8.4e+08	5.9e+06/ 8.9e+04	8.5e+02/ 7.8e+02	2.7e+04/ 1.3e+04	8.6e+02/ 7.8e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.6e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.0e+06/ 1.2e+05	7.9e+02/ 3.9e+02	1.9e+04/ 7.7e+03	7.9e+02/ 3.9e+02	1.4e+07/ 3.5e+05
sumImpurity	F	2.6e+08/ 1.6e+08	7.6e+08/7.8e+08	7.5e+06/ 5.7e+05	7.6e+02/ 7.3e+02	3.6e+04/ 1.3e+04	7.7e+02/ 7.3e+02	1.5e+07/ 2.0e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08/9.0e+08	7.7e+06/ 3.5e+05	9.3e+02/ 8.1e+02	3.8e+04/ 1.4e+04	9.3e+02/ 8.1e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.5e+08/ 1.6e+08	6.4e+08/7.7e+08	6.2e+06/ 2.5e+05	7.5e+02/8.7e+02	3.7e+04/ 1.4e+04	7.6e+02/8.7e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.6e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.2e+06/ 1.1e+05	8.3e+02/ 2.3e+02	1.7e+04/ 7.8e+03	8.3e+02/ 2.3e+02	1.4e+07/ 3.2e+05
Total		2.0e+10/1.5e+10	4.0e+10/6.8e+10	5.4e+08/2.9e+07	1.0e+05/4.6e+04	2.1e+06/2.1e+06	1.0e+05/4.4e+04	1.1e+09/2.9e+07

Table 14: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
SumArea	F	6/15	60%	0.0372 $\pm$ 0.00001	0.0713 $\pm$ 0.00015	0.0192$\pm$0.00003	1.94$\times$	1.91$\times$	3.71$\times$
computeWidths	M	14/15	7%	0.0295$\pm$0.00004	0.0301 $\pm$ 0.00003	0.0438 $\pm$ 0.00004	0.67 $\times$	1.02 $\times$	0.69 $\times$
count styled	F	3/15	80%	0.0360 $\pm$ 0.00001	0.0710 $\pm$ 0.00125	0.0116$\pm$0.00001	3.09$\times$	1.97$\times$	6.10$\times$
find max Bottom	F	5/15	67%	0.0362 $\pm$ 0.00013	0.0706 $\pm$ 0.00001	0.0188$\pm$0.00004	1.93$\times$	1.95$\times$	3.76$\times$
scaleLayout	M	15/15	0%	0.0286$\pm$0.00003	0.0300 $\pm$ 0.00004	0.0346 $\pm$ 0.00089	0.83 $\times$	1.05 $\times$	0.87 $\times$
sumTextWidth	F	3/15	80%	0.0358 $\pm$ 0.00001	0.0716 $\pm$ 0.00019	0.0117$\pm$0.00001	3.07$\times$	2.00$\times$	6.14$\times$
Total				0.2033	0.3446	0.1397	1.46$\times$	1.70$\times$	2.47$\times$
Geomean						1.64$\times$	1.58$\times$	2.60$\times$	

Table 15: Per-pass PAPI counters for DomTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.9e+08/ 9.7e+07	2.9e+08/4.9e+08	2.9e+06/ 5.7e+05	4.3e+02/ 5.0e+01	1.7e+05/ 1.2e+04	4.4e+02/ 5.1e+01	1.3e+07/ 2.0e+06
computeWidths	M	6.5e+07/1.4e+08	1.4e+08/4.6e+08	1.2e+06/1.5e+06	4.9e+04/ 4.5e+04	5.8e+03/1.2e+04	1.2e+03/1.5e+04	1.2e+06/ 4.3e+05
count styled	F	1.8e+08/ 5.9e+07	2.6e+08/2.6e+08	2.3e+06/ 2.5e+04	7.9e+02/ 8.6e+01	3.0e+05/ 5.0e+03	7.9e+02/ 9.5e+01	1.4e+07/ 6.5e+05
find max Bottom	F	1.8e+08/ 9.5e+07	2.5e+08/4.1e+08	2.7e+06/ 1.4e+06	6.3e+02/ 5.7e+01	1.4e+05/ 9.7e+03	6.3e+02/ 5.8e+01	1.3e+07/ 1.4e+06
scaleLayout	M	6.0e+07/8.8e+07	8.8e+07/2.9e+08	1.0e+06/ 7.0e+05	6.7e+04/ 6.6e+04	1.5e+04/ 8.3e+03	1.4e+03/1.6e+04	1.2e+06/ 4.5e+05
sumTextWidth	F	1.8e+08/ 5.9e+07	2.0e+08/2.4e+08	2.5e+06/ 2.0e+04	7.4e+02/ 8.0e+01	4.2e+05/ 4.2e+03	7.4e+02/ 8.4e+01	1.3e+07/ 6.5e+05
Total		8.6e+08/5.3e+08	1.2e+09/2.2e+09	1.3e+07/4.2e+06	1.2e+05/1.1e+05	1.1e+06/5.1e+04	5.2e+03/3.1e+04	5.6e+07/5.5e+06

Table 16: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
Find nearest Neighbour	F	13/17	24%	0.0292 $\pm$ 0.00002	0.0391 $\pm$ 0.00008	0.0210$\pm$0.00004	1.39$\times$	1.34$\times$	1.87$\times$
countInRange tight_box	F	13/17	24%	0.0294 $\pm$ 0.00073	0.0399 $\pm$ 0.00002	0.0197$\pm$0.00001	1.49$\times$	1.36$\times$	2.02$\times$
photonMappingBenchmark	F	12/17	29%	0.0439 $\pm$ 0.00015	0.0473 $\pm$ 0.00001	0.0408$\pm$0.00002	1.08 $\times$	1.08 $\times$	1.16$\times$
pointCloudNeighborhood	F	11/17	35%	0.0264 $\pm$ 0.00002	0.0383 $\pm$ 0.00013	0.0165$\pm$0.00000	1.60$\times$	1.45$\times$	2.32$\times$
sumMassInRange	F	14/17	18%	0.0286 $\pm$ 0.00001	0.0399 $\pm$ 0.00002	0.0197$\pm$0.00001	1.45$\times$	1.39$\times$	2.02$\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0304 $\pm$ 0.00002	0.0390 $\pm$ 0.00001	0.0209$\pm$0.00000	1.46$\times$	1.28$\times$	1.87$\times$
Total				0.1879	0.2435	0.1386	1.36$\times$	1.30$\times$	1.76$\times$
Geomean						1.40$\times$	1.31$\times$	1.84$\times$	

Table 17: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.5e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.9e+06/ 1.8e+06	8.6e+01/ 7.9e+01	4.5e+03 /1.7e+04	8.7e+01/ 7.9e+01	7.6e+06/ 1.6e+06
countInRange tight_box	F	1.5e+08/ 1.0e+08	4.0e+08 /5.5e+08	2.1e+06/ 2.0e+06	1.7e+02 /1.8e+02	3.0e+03 /1.6e+04	1.7e+02 /1.8e+02	7.6e+06/ 1.6e+06
photonMappingBenchmark	F	2.2e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.2e+06 /2.5e+06	2.5e+02 /4.4e+02	3.2e+03 /1.8e+04	2.5e+02 /4.4e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.3e+08/ 8.4e+07	3.3e+08 /4.8e+08	2.7e+06/ 1.3e+06	1.7e+02/ 1.2e+02	4.4e+03 /1.5e+04	1.8e+02/ 1.2e+02	7.8e+06/ 1.4e+06
sumMassInRange	F	1.5e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.5e+06 /3.7e+06	6.8e+01 /1.4e+02	4.0e+03 /2.0e+04	6.8e+01 /1.4e+02	7.6e+06/ 1.7e+06
twoPointCorrelation bin_8_16	F	1.5e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.2e+06/ 6.3e+05	1.3e+02/ 9.3e+01	2.8e+03 /1.3e+04	1.3e+02/ 9.2e+01	7.6e+06/ 1.2e+06
Total		9.5e+08/7.0e+08	3.1e+09/4.0e+09	1.5e+07/1.2e+07	8.8e+02/1.1e+03	2.2e+04/9.9e+04	8.8e+02/1.0e+03	4.4e+07/8.9e+06

Table 18: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
reduction	F	2/11	82%	0.0296 $\pm$ 0.00012	0.1330 $\pm$ 0.00021	0.0050$\pm$0.00001	5.91$\times$	4.50$\times$	26.59$\times$
Total				0.0296	0.1330	0.0050	5.91$\times$	4.50$\times$	26.59$\times$
Geomean						5.91$\times$	4.50$\times$	26.59$\times$	

Table 19: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	2.3e+02/ 4.3e+01	2.8e+05/ 7.0e+03	2.3e+02/ 4.3e+01	1.2e+07/ 1.1e+06
Total		1.5e+08/2.5e+07	9.0e+07/1.0e+08	1.5e+06/4.6e+04	2.3e+02/4.3e+01	2.8e+05/7.0e+03	2.3e+02/4.3e+01	1.2e+07/1.1e+06

Table 20: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
add1 List	M	2/2	0%	0.2562$\pm$0.00214	<i>OOM</i>	0.3395 $\pm$ 0.00266	0.75 $\times$	–	–
length List	F	1/2	50%	0.0419 $\pm$ 0.00001	<i>OOM</i>	0.0215$\pm$0.00120	1.95$\times$	–	–
sumList	F	2/2	0%	0.0441$\pm$0.00001	<i>OOM</i>	0.0455 $\pm$ 0.05984	0.97 $\times$	–	–
sumList tail recursive	F	2/2	0%	0.0444$\pm$0.00002	<i>OOM</i>	0.0449 $\pm$ 0.00001	0.99 $\times$	–	–
Total				0.3867	–	0.4514	0.86 $\times$	–	–
Geomean						1.09 $\times$	–	–	

Table 21: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.7e+08	1.9e+09 /2.8e+09	7.9e+06/ 1.5e+06	4.2e+05 /6.0e+05	1.3e+04 /5.5e+04	5.9e+03 /9.9e+03	9.2e+06/ 7.7e+06
length List	F	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.3e+05/ 9.8e+04	1.1e+02/ 7.5e+01	2.6e+03 /1.4e+04	1.1e+02/ 7.5e+01	1.3e+07/ 1.1e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	6.6e+05/ 5.5e+05	9.9e+01/ 5.3e+01	3.0e+03 /6.8e+04	9.9e+01/ 5.6e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.1e+05 /5.7e+05	1.1e+02 /1.7e+02	2.9e+03 /5.8e+04	1.1e+02 /1.7e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.2e+06/2.7e+06	4.2e+05/6.0e+05	2.2e+04/2.0e+05	6.2e+03/1.0e+04	4.9e+07/3.3e+07

Table 22: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
add1Tree	M	3/3	0%	0.0454 $\pm$ 0.00015	0.0610 $\pm$ 0.00060	0.0427$\pm$0.00035	1.06 $\times$	1.34$\times$	1.43$\times$
sumTree	F	3/3	0%	0.0091$\pm$0.00001	0.0406 $\pm$ 0.00002	0.0114 $\pm$ 0.00001	0.80 $\times$	4.44$\times$	3.57$\times$
sumTree TailRec	F	3/3	0%	0.0086$\pm$0.00001	0.0402 $\pm$ 0.00001	0.0111 $\pm$ 0.00001	0.77 $\times$	4.67$\times$	3.61$\times$
Total				0.0631	0.1418	0.0652	0.97 $\times$	2.25$\times$	2.18$\times$
Geomean						0.87 $\times$	3.03$\times$	2.64$\times$	

Table 23: Per-pass PAPI counters for **MonoTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.2e+05 /4.0e+05	3.2e+04 /4.4e+04	2.0e+03 /5.1e+03	1.4e+03 /1.9e+03	5.1e+05/ 4.6e+05
sumTree	F	4.6e+07 /5.8e+07	2.2e+08 /2.6e+08	1.1e+04 /3.4e+04	1.6e+02/ 1.4e+02	6.9e+02 /4.0e+03	1.5e+02/ 1.5e+02	7.6e+05/ 6.1e+05
sumTree TailRec	F	4.4e+07 /5.6e+07	1.9e+08 /2.2e+08	6.5e+03 /2.0e+04	1.8e+02 /1.9e+02	7.5e+02 /4.0e+03	1.8e+02 /1.9e+02	7.5e+05/ 6.0e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.4e+05/4.5e+05	3.2e+04/4.4e+04	3.4e+03/1.3e+04	1.7e+03/2.2e+03	2.0e+06/1.7e+06

Table 24: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countLargeItems	F	3/5	40%	0.0138 $\pm$ 0.00000	0.0448 $\pm$ 0.00002	0.0108$\pm$0.00002	1.27 $\times$	3.26 $\times$	4.14 $\times$
countMarked	F	3/5	40%	0.0136 $\pm$ 0.00000	0.0450 $\pm$ 0.00002	0.0109$\pm$0.00002	1.25 $\times$	3.30 $\times$	4.14 $\times$
countSurvivors	F	4/5	20%	0.0146 $\pm$ 0.00001	0.0450 $\pm$ 0.00113	0.0130$\pm$0.00001	1.12 $\times$	3.09 $\times$	3.46 $\times$
deadBytes	F	4/5	20%	0.0143 $\pm$ 0.00000	0.0444 $\pm$ 0.00002	0.0125$\pm$0.00001	1.14 $\times$	3.11 $\times$	3.55 $\times$
liveBytes	F	4/5	20%	0.0147 $\pm$ 0.00001	0.0437 $\pm$ 0.00002	0.0125$\pm$0.00001	1.17 $\times$	2.98 $\times$	3.49 $\times$
sumObjIds	F	3/5	40%	0.0134 $\pm$ 0.00000	0.0436 $\pm$ 0.00093	0.0105$\pm$0.00002	1.27 $\times$	3.26 $\times$	4.13 $\times$
sweepUnmarked	M	5/5	0%	0.0727$\pm$0.00008	0.0879 $\pm$ 0.00009	0.0798 $\pm$ 0.00009	0.91 $\times$	1.21 $\times$	1.10 $\times$
totalHeapSize	F	3/5	40%	0.0133 $\pm$ 0.00001	0.0435 $\pm$ 0.00029	0.0104$\pm$0.00011	1.29 $\times$	3.27 $\times$	4.20 $\times$
touchHotObjects	M	5/5	0%	0.0718$\pm$0.00005	0.0875 $\pm$ 0.00005	0.0804 $\pm$ 0.00016	0.89 $\times$	1.22 $\times$	1.09 $\times$
Total				0.2420	0.4854	0.2408	1.01 $\times$	2.01 $\times$	2.02 $\times$
Geomean						1.14 $\times$	2.57 $\times$	2.92 $\times$	

Table 25: Per-pass PAPI counters for **ObjectGraph** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	7.0e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.1e+06/ 5.1e+04	3.6e+02/ 1.5e+02	3.0e+03 /3.7e+03	3.7e+02/ 1.6e+02	2.8e+06/ 6.3e+05
countMarked	F	6.9e+07/ 5.5e+07	2.6e+08 /2.8e+08	1.2e+06/ 5.1e+04	1.5e+02/ 1.3e+02	3.3e+03 /4.0e+03	1.6e+02/ 1.4e+02	2.8e+06/ 6.4e+05
countSurvivors	F	7.4e+07/ 6.6e+07	3.1e+08 /3.8e+08	1.0e+06/ 7.4e+04	2.8e+02/ 2.2e+02	2.5e+03 /5.2e+03	2.8e+02/ 2.2e+02	2.8e+06/ 1.2e+06
deadBytes	F	7.2e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.7e+05/ 5.9e+04	2.8e+02/ 1.3e+02	2.3e+03 /5.4e+03	2.9e+02/ 1.4e+02	2.7e+06/ 1.2e+06
liveBytes	F	7.4e+07/ 6.3e+07	2.7e+08 /3.5e+08	6.9e+05/ 5.8e+04	3.2e+02/ 1.8e+02	2.0e+03 /5.4e+03	3.3e+02/ 1.8e+02	2.7e+06/ 1.2e+06
sumObjIds	F	6.8e+07/ 5.3e+07	2.5e+08 /2.7e+08	1.3e+06/ 8.0e+04	2.4e+02/ 1.2e+02	3.1e+03 /4.4e+03	2.4e+02/ 1.3e+02	2.8e+06/ 6.1e+05
sweepUnmarked	M	1.9e+08 /2.3e+08	5.5e+08 /8.8e+08	9.8e+05/ 4.2e+05	1.0e+05 /1.4e+05	4.7e+03 /1.1e+04	3.1e+03 /8.8e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.7e+07/ 5.2e+07	2.5e+08 /2.7e+08	1.3e+06/ 1.1e+05	1.9e+02/ 1.4e+02	3.0e+03 /3.7e+03	2.0e+02/ 1.4e+02	2.8e+06/ 6.4e+05
touchHotObjects	M	1.8e+08 /2.3e+08	5.7e+08 /9.0e+08	8.1e+05/ 4.6e+05	1.2e+05/ 1.2e+05	5.1e+03 /1.1e+04	4.4e+03 /8.1e+03	2.0e+06/ 1.0e+06
Total		8.7e+08/8.7e+08	3.0e+09/4.0e+09	9.2e+06/1.4e+06	2.3e+05/2.6e+05	2.9e+04/5.3e+04	9.4e+03/1.8e+04	2.3e+07/8.2e+06

Table 26: Per-pass performance for **PiecewiseFunctions**, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
addConstPW	M	8/8	0%	0.1196$\pm$0.00110	0.1297 $\pm$ 0.00164	0.1307 $\pm$ 0.00011	0.92 $\times$	1.08 $\times$	0.99 $\times$
autorefineMaxLevel	F	4/8	50%	0.0212$\pm$0.00001	0.0460 $\pm$ 0.00136	0.0212 $\pm$ 0.00001	1.00 $\times$	2.17 $\times$	2.17 $\times$
compressMass	F	3/8	62%	0.0202 $\pm$ 0.00000	0.0468 $\pm$ 0.00002	0.0116$\pm$0.00009	1.74 $\times$	2.31 $\times$	4.03 $\times$
diffPW	M	8/8	0%	0.1191$\pm$0.00011	0.1321 $\pm$ 0.00096	0.1286 $\pm$ 0.00123	0.93 $\times$	1.11 $\times$	1.03 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0222 $\pm$ 0.00001	0.0487 $\pm$ 0.00002	0.0188$\pm$0.00002	1.18 $\times$	2.20 $\times$	2.58 $\times$
norm2Estimate	F	5/8	38%	0.0257 $\pm$ 0.00001	0.0556 $\pm$ 0.00157	0.0184$\pm$0.00005	1.39 $\times$	2.16 $\times$	3.02 $\times$
pmapCutHistogram	F	4/8	50%	0.0327 $\pm$ 0.00001	0.0570 $\pm$ 0.00002	0.0128$\pm$0.00001	2.57 $\times$	1.74 $\times$	4.47 $\times$
truncateTolViolations	F	3/8	62%	0.0207 $\pm$ 0.00001	0.0474 $\pm$ 0.00001	0.0116$\pm$0.00001	1.78 $\times$	2.29 $\times$	4.08 $\times$
Total				0.3815	0.5633	0.3538	1.08 $\times$	1.48 $\times$	1.59 $\times$
Geomean						1.35 $\times$	1.81 $\times$	2.44 $\times$	

Table 27: Per-pass PAPI counters for **PiecewiseFunctions** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.8e+08 /3.3e+08	5.7e+08 /1.2e+09	1.2e+06 /2.4e+06	3.3e+05/ 2.0e+05	9.6e+03 /2.5e+04	6.8e+03 /2.2e+04	4.5e+06/ 1.8e+06
autorefineMaxLevel	F	1.1e+08 /1.1e+08	2.5e+08 /3.7e+08	2.5e+06/ 2.0e+05	4.9e+02/ 8.6e+01	9.0e+03/ 6.3e+03	4.9e+02/ 8.9e+01	6.1e+06/ 9.0e+05
compressMass	F	1.0e+08/ 5.9e+07	2.5e+08 /2.8e+08	2.2e+06/ 3.0e+04	6.0e+02/ 4.1e+02	1.9e+04/ 4.9e+03	6.0e+02/ 4.1e+02	6.2e+06/ 6.1e+05
diffPW	M	2.7e+08 /3.2e+08	5.9e+08 /1.3e+09	8.5e+05 /2.3e+06	1.5e+05 /1.9e+05	9.5e+03 /2.5e+04	9.7e+03 /2.3e+04	4.5e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.6e+07	3.3e+08 /4.9e+08	2.4e+06/ 3.4e+05	2.8e+02/ 1.4e+02	1.3e+04/ 8.9e+03	2.8e+02/ 1.4e+02	6.3e+06/ 1.6e+06
norm2Estimate	F	1.3e+08/ 9.4e+07	3.4e+08 /4.4e+08	1.2e+06/ 5.0e+05	2.2e+02/ 1.1e+02	1.1e+04/ 9.4e+03	2.2e+02/ 1.1e+02	6.1e+06/ 1.6e+06
pmapCutHistogram	F	1.7e+08/ 6.5e+07	2.8e+08 /3.7e+08	1.6e+06/ 6.6e+04	7.0e+02/ 3.5e+02	9.1e+04/ 7.6e+03	7.0e+02/ 3.6e+02	4.8e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.9e+07	2.5e+08 /2.9e+08	2.2e+06/ 3.0e+04	5.5e+02/ 1.4e+02	1.5e+04/ 4.8e+03	5.5e+02/ 1.4e+02	6.3e+06/ 6.3e+05
Total		1.3e+09/1.1e+09	2.8e+09/4.7e+09	1.4e+07/5.8e+06	4.9e+05/4.0e+05	1.8e+05/9.2e+04	1.9e+04/4.7e+04	4.5e+07/1.0e+07

Table 28: Per-pass performance for **TernaryTree**, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
add 1 tree	M	5/5	0%	0.0830 $\pm$ 0.00008	0.0945 $\pm$ 0.00060	0.0779$\pm$0.00025	1.07 $\times$	1.14 $\times$	1.21 $\times$
sum tree	F	5/5	0%	0.0277 $\pm$ 0.00009	0.0501 $\pm$ 0.00002	0.0257$\pm$0.00009	1.08 $\times$	1.81 $\times$	1.95 $\times$
Total				0.1107	0.1446	0.1035	1.07 $\times$	1.31 $\times$	1.40 $\times$
Geomean						1.07 $\times$	1.43 $\times$	1.54 $\times$	

Table 29: Per-pass PAPI counters for **TernaryTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.6e+05 /1.3e+06	6.0e+04 /1.5e+05	3.6e+03 /1.1e+04	2.2e+03 /2.6e+03	1.5e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	9.0e+04 /3.6e+05	5.9e+01 /9.1e+01	8.2e+02 /9.2e+03	6.2e+01 /9.1e+01	1.8e+06/ 1.3e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	5.5e+05/1.6e+06	6.0e+04/1.5e+05	4.4e+03/2.0e+04	2.3e+03/2.7e+03	3.3e+06/2.4e+06

Table 30: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autocompleteTopKProxy	F	5/10	50%	0.0148 $\pm$ 0.00001	0.0271 $\pm$ 0.00014	0.0086$\pm$0.00004	1.72 $\times$	1.84 $\times$	3.15 $\times$
countLazyNodes	F	4/10	60%	0.0133 $\pm$ 0.00050	0.0274 $\pm$ 0.00001	0.0066$\pm$0.00001	2.00 $\times$	2.06 $\times$	4.13 $\times$
countTerminals	F	3/10	70%	0.0126 $\pm$ 0.00001	0.0254 $\pm$ 0.00001	0.0058$\pm$0.00000	2.16 $\times$	2.02 $\times$	4.37 $\times$
decayTrieStats	M	10/10	0%	0.0744$\pm$0.00108	0.0802 $\pm$ 0.00104	0.0855 $\pm$ 0.00010	0.87 $\times$	1.08 $\times$	0.94 $\times$
resetTraversalState	M	10/10	0%	0.0753$\pm$0.00012	0.0814 $\pm$ 0.00010	0.0840 $\pm$ 0.00016	0.90 $\times$	1.08 $\times$	0.97 $\times$
sumPrefixFreq	F	3/10	70%	0.0128 $\pm$ 0.00001	0.0258 $\pm$ 0.00015	0.0057$\pm$0.00006	2.24 $\times$	2.01 $\times$	4.51 $\times$
sumSubtreeHints	F	3/10	70%	0.0128 $\pm$ 0.00001	0.0258 $\pm$ 0.00001	0.0058$\pm$0.00001	2.22 $\times$	2.01 $\times$	4.47 $\times$
Total				0.2160	0.2931	0.2021	1.07 $\times$	1.36 $\times$	1.45 $\times$
Geomean						1.62 $\times$	1.67 $\times$	2.70 $\times$	

Table 31: Per-pass PAPI counters for **Trie** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.5e+07/ 4.3e+07	1.3e+08 /2.1e+08	1.5e+06/ 3.8e+05	6.2e+01/ 5.0e+01	4.2e+04/ 4.6e+03	6.9e+01/ 5.0e+01	3.7e+06/ 7.5e+05
countLazyNodes	F	6.7e+07/ 3.4e+07	1.3e+08 /1.6e+08	1.4e+06/ 2.1e+04	7.8e+01 /8.0e+01	1.6e+04/ 3.3e+03	8.6e+01/ 8.0e+01	4.0e+06/ 5.6e+05
countTerminals	F	6.4e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.4e+06/ 1.0e+04	9.5e+01/ 6.3e+01	2.8e+04/ 2.3e+03	1.0e+02/ 6.3e+01	4.0e+06/ 2.1e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	8.7e+05 /1.3e+06	8.7e+04 /1.1e+05	6.0e+03 /2.1e+04	3.5e+03 /2.7e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.1e+08	2.8e+08 /7.0e+08	9.4e+05/ 7.0e+05	8.6e+04 /1.5e+05	6.3e+03 /1.3e+04	2.1e+03 /1.4e+04	3.0e+06/ 9.0e+05
sumPrefixFreq	F	6.5e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.4e+04	8.5e+01/ 5.4e+01	2.2e+04/ 2.5e+03	9.0e+01/ 5.4e+01	4.0e+06/ 2.1e+05
sumSubtreeHints	F	6.5e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.4e+04	1.2e+02/ 5.7e+01	5.2e+04/ 2.5e+03	1.2e+02/ 5.7e+01	3.9e+06/ 2.1e+05
Total		6.6e+08/5.9e+08	1.2e+09/2.2e+09	9.2e+06/2.4e+06	1.7e+05/2.6e+05	1.7e+05/4.9e+04	6.0e+03/4.2e+04	2.6e+07/4.0e+06